



OPEN Machine learning combined with multi-omics to identify immune-related LncRNA signature as biomarkers for predicting breast cancer prognosis

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This study developed an immune-related long non-coding RNAs (lncRNAs)-based prognostic signature by integrating multi-omics data and machine learning algorithms to predict survival and therapeutic responses in breast cancer patients. Utilizing transcriptomic and gene expression data from TCGA and GEO databases, 72 immune-related lncRNAs were identified through weighted gene co-expression network analysis (WGCNA) and ImmLncRNA algorithms. The model was further optimized using 101 combinations of 10 machine learning approaches, ultimately constructing an immune-related lncRNA signature (IRLS) scoring system comprising nine key lncRNAs. Validated across 17 independent cohorts, the model demonstrated that high-risk patients had significantly shorter overall survival (OS) ($P < 0.05$), with predictive performance surpassing 95 published models ($P < 0.05$). Additionally, the IRLS score predicted responses to paclitaxel chemotherapy, and the low-risk group exhibited higher immune cell infiltration ($P < 0.05$), showing significant negative correlations with CD8A, PD-L1, tumor mutational burden (TMB), and neoantigen load (NAL). In immune checkpoint inhibitor (ICI) treatment cohorts, low IRLS scores were associated with improved response rates to atezolizumab. Our findings suggest that the IRLS model serves as a novel biomarker for prognostic stratification and personalized therapeutic decision-making in breast cancer.

Keywords Breast cancer, Long non-coding RNA, Immune microenvironment, Machine learning, Prognostic model, Treatment prediction

The incidence of breast cancer has been increasing over the past few decades, replacing lung cancer as the most common malignant tumor globally to date¹. Breast cancer can be classified into four distinct molecular subtypes based on estrogen receptor (ER) expression, progesterone receptor (PR) expression, and human epidermal growth factor receptor 2 (HER2) gene amplification. Different subtypes have different risk profiles and optimal treatment strategies². Over the past few decades, significant progress has been made in the treatment of breast cancer. Surgery is generally the first choice for patients with breast cancer, but it still requires a comprehensive evaluation considering factors such as molecular subtype, TNM staging, and age to select the most appropriate neoadjuvant therapy, chemotherapy, radiotherapy, and targeted therapy³. Early detection and treatment often result in good prognosis for patients. However, breast cancer has complex pathological subtypes, unique

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biological microenvironments, and tumor heterogeneity, making it impossible to achieve a complete cure or eradication at present. Recurrence and metastasis after breast cancer treatment remain the primary cause of patient death, with an estimated 20–30% of breast cancer patients experiencing metastasis following diagnosis and initial tumor treatment⁴. The metastasis of breast cancer is a complex dynamic process involving multiple systems throughout the body, including the invasion of tumor cells, their infiltration into blood vessels and the emergence of circulating tumor cells, extravasation into tissues, and ultimately the formation of micro-metastases and large-scale metastases⁵. Therefore, early detection, early diagnosis, and early treatment are of great significance for breast cancer patients. Currently, clinical tests such as 21-gene testing, 70-gene testing, 50-gene recurrence risk, and 12-gene testing are primarily used for early-stage breast cancer patients who are hormone receptor-positive and HER2-negative^{6–9}. Due to the significant differences in molecular characteristics, biological behavior, and risk profiles among various subtypes of breast cancer, clinical management and personalized therapy remain a significant challenge.

Long non-coding RNAs (lncRNAs) are a class of RNA molecules longer than 200 nucleotides that lack the ability to encode proteins¹⁰. Initially, they were considered as “noise” produced during genome transcription with no biological function. Later research revealed their crucial regulatory roles in various physiological and pathological processes. Abnormal expression of lncRNAs are closely associated with tumor development, metastasis, and chemotherapy resistance. In particular, its role in breast cancer progression can be divided into two categories: promoting and inhibiting breast cancer development. The mechanisms typically involve affecting the proliferation and apoptosis of breast cancer cells, drug resistance, and invasiveness¹¹. Although many lncRNAs play a vital role in breast cancer, there is a lack of immune-related lncRNA to predict breast cancer prognosis. Therefore, in the era of precision medicine, developing new prognostic factors and predictive methods for breast cancer can provide essential guidance for predicting breast cancer outcomes and individualized treatment for breast cancer patients.

In this study, based on The Cancer Genome Atlas (TCGA) and the gene expression omnibus (GEO), we used bioinformatics to screen for lncRNAs related to immunity and constructed an lncRNAs model associated with breast cancer prognosis. Additionally, the study analyzed differences in immune cell infiltration and immune checkpoints across different risk groups, aiming to effectively identify potential targets for high-risk patients, thereby tailoring more personalized treatment plans for each patient.

Materials and methods

Datasets and preprocessing

In our study, a total of 2,607 breast cancer patients were included, and data from 17 public databases were accessed. Among these, transcriptome sequencing data and clinical information for 1,108 breast cancer tissues were downloaded through TCGA (<https://portal.gdc.cancer.gov/>) to construct our signature. Additionally, gene chip data and clinical information for 1,315 patients were obtained from the GEO (<https://www.ncbi.nlm.nih.gov/geo/>) database. The GSE42568 ($n=88$), GSE58812 ($n=107$), GSE103091 ($n=238$), GSE20711 ($n=88$), GSE20713 ($n=104$), GSE16446 ($n=102$) dataset contains patient overall survival (OS) and disease-free survival (DFS) data, while the GSE9195 ($n=77$), GSE20685 ($n=56$), GSE21653 ($n=211$), GSE61304 ($n=244$) dataset includes complete DFS data. All datasets were generated using the GPL570 platform. In addition, for the drug-related datasets, we separately collected treatment information from 112 patients across six datasets from GSE50948, GSE16179, GSE22513, GSE28844, GSE66305, and GSE33568. These datasets were used to predict the performance of immune-related lncRNA signature (IRLS) in evaluating different breast cancer treatment regimens. Furthermore, we assessed the predictive performance of IRLS on immune checkpoint inhibitor (ICI) therapy efficacy using data from 82 patients in the GSE157284 dataset.

The transcriptomics data downloaded from the TCGA database were converted to transcripts per million (TPM) format. GEO data generated from the GPL570 platform were processed using GC robust multichip average method (GC-RMA). We aligned the TCGA data with genomic annotation files obtained from GENCODE (Homo sapiens GRCh38), resulting in 16,853 lncRNAs. Using the SeqMap tool, we re-annotated the probe sequences from GPL570, ultimately obtaining 3,419 lncRNAs¹².

Estimated immune cell infiltration

We used the single-sample gene set enrichment analysis (ssGSEA) algorithm to calculate the 28 immune cell infiltration scores for TCGA-BRCA samples¹³. Additionally, we evaluated the reliability and extrapolation of our results using five algorithms from the IOBR package (TIMER, quanTIseq, MCP-counter, xCell, EPIC) and ESTIMATE immune scores.

Consensus clustering

According to the immune score calculated by ssGSEA, we performed unsupervised consensus clustering on the TCGA-BRCA dataset using the ConsensusClusterPlus package¹⁴. The consensus heatmap, cumulative distribution function (CDF), proportion of ambiguous clustering (PAC), and Number of Clusters (Nbclust) algorithm were used to screen for the most robust cluster configuration¹⁵.

WGCNA analyse

Using the TCGA-BRCA dataset, the weighted gene co-expression network analysis (WGCNA) package was employed to explore the co-expression network of lncRNA. The optimal soft threshold β was calculated using the pickSoftThreshold function. Additionally, the blockwiseModules function was applied to identify co-expression modules of lncRNAs. The minimum number of lncRNAs in a module was set to 30, with 0.25 as the standard threshold for merging modules, and different colors were used to distinguish various lncRNA co-expression modules. Each lncRNA's association with clinical significance was measured by gene significance (GS), and the

module membership (MM) refers to the correlation between lncRNAs and gene expression profiles. lncRNA with $|GS| > 0.2$ and $|MM| > 0.8$ were selected as candidate immune-related lncRNAs.

ImmLnc analysis

We identified potential lncRNAs regulators involved in immune-related pathways using the TCGA-BRCA cohort and the ImmLncRNA algorithm^{16,17}. First, we calculated a ranking score (RS) based on the output of partial correlation analysis (PCC), which used lncRNA and gene expression data along with tumor purity for corresponding samples. Subsequently, all genes were ranked by their RS scores, and immune genes in each pathway were mapped to the sorted list of genes. Based on gene set enrichment analysis (GSEA), we obtained enrichment scores (ES) and p-values. Finally, we integrated the p-values and ES scores into lncRES scores and selected lncRNA-pathway pairs with $|lncRES| > 0.995$ and $FDR < 0.05$ as significantly regulated relationships.

Machine learning-based construction of a consensus immune-related lncRNA signature

We performed a univariate Cox analysis on the intersection of immune-related lncRNA identified by WGCNA and ImmLnc. We identified 15 immune-related lncRNA for subsequent machine learning evaluation to generate an IRLS. To enhance the reliability and extrapolation of our results, we used 10 machine learning algorithms (random forest (RSF), elastic network (Enet), stepwise Cox (StepCox), CoxBoost, partial least squares regression for Cox (plsRcox), supervised principal components (superpc), generalized boosted regression models (GBM), survival support vector machine (survivalsvm), Ridge, and least absolute shrinkage and selection operator (Lasso)) to construct 101 algorithm combinations to find the optimal patient prognosis model. We used TCGA-BRCA as the training set of the model, and GSE58812, GSE103091, GSE42568, GSE20711, GSE16446 and GSE20713 as the validation set for the development of the model. We selected the algorithm combination with the highest average C-Index as the model for screening IRLS.

Evaluation of the IRLS model

To evaluate the ability of the IRLS model to distinguish survival outcomes, we used the timeROC package to calculate the AUC values for 1-, 3-, and 5-year OS in patients. Subsequently, we used the survival package to compute the C-index values across different datasets to assess the accuracy of the model in predicting patient prognosis risks. Additionally, we employed the survex and risksetROC algorithms to calculate the integrated Bier score (IBS) and integrated AUC (iAUC) for different datasets to evaluate the model's accuracy in predicting survival risks. To compare the accuracy of IRLS with other clinical features in predicting survival outcomes, we used the compareC package to calculate the C-index values for 8 different datasets, including both clinical features and IRLS, and compared whether there were significant differences in these C-index values compared to IRLS. Subsequently, in the TCGA-BRCA dataset, we systematically evaluated a combined prognostic metric integrating AJCC Stage and IRLS to determine whether this multimodal approach could surpass single-indicator models (AJCC Stage alone or IRLS alone) in survival prediction accuracy. This comprehensive analysis specifically investigated the potential of the AJCC Stage-IRLS integration to outperform existing clinical prognostic markers in terms of predictive performance.

Assessment of gene expression-derived predictive models for breast cancer

To evaluate the robustness of our constructed IRLS prediction model, we collected 95 prognostic models of gene expression from published literature in Pubmed. First, we calculated the risk scores of each model on eight datasets (TCGA-BRCA, GSE58812, GSE103091, GSE42568, GSE20711, GSE16446, GSE20713, and Meta-Cohort) using univariate Cox regression to assess their impact on OS in breast cancer patients. Subsequently, we computed the C-index for each model and compared the predictive performance of different prognostic models with IRLS on OS using the compareC algorithm.

Predictive value of IRLS score for different treatment regimens

Studies have shown that lncRNA may regulate the sensitivity of tumor cells to chemotherapeutic drugs (such as Paclitaxel) through certain signaling pathways^{18–20}. To explore the predictive value of our constructed IRLS scoring model for patient treatment efficacy, we collected six independent GEO datasets (GSE50948, GSE16179, GSE22513, GSE28844, GSE66305, and GSE33568) to investigate the predictive value of IRLS scores for different drug treatments.

Impact of IRLS on immune checkpoint inhibitor therapy

To explore the predictive power of IRLS in ICI treatment for breast cancer patients, we used the ssGSEA algorithm to calculate the immune cell infiltration status of patients in the TCGA-BRCA and Meta-GEO datasets, displaying it as heatmaps. Additionally, we investigated the correlation between IRLS and different immune cells. Subsequently, we conducted exploratory analyses on the correlation between IRLS and CD8A and PD-L1 using the TCGA-BRCA dataset and the GSE20711 dataset, which provides more comprehensive clinical information. We also studied the predictive performance of IRLS scores in atezolizumab treatment response using the GSE157284 dataset.

Statistical analysis

For categorical variables and continuous variables, we employed chi-square analysis and two-tailed tests respectively. The optimal cutoff value was determined using the survminer algorithm. Batch effects were removed through the ComBat R package, followed by integration of all datasets to establish the Meta-Cohort dataset. Correlations between variables were evaluated using Pearson's correlation coefficients. To eliminate technical

biases, we performed z-score standardization on genes across all cohorts. All data processing, statistical analyses, and graphical visualizations were implemented using R software (version 4.4.1).

Results

Identifying immune infiltration subtypes via consensus clustering

We calculated the enrichment scores of 28 immune cell types in the TCGA-BRCA cohort using ssGSEA. Subsequently, we performed unsupervised consensus clustering based on the area under the CDF curve and consensus heatmaps to determine the number of clusters. When k was set to 2, the heatmap clearly divided breast cancer patients into two clusters with distinct expression patterns (Fig. 1A, B). Additionally, the optimal number of clusters identified by PAC and Nbclust algorithms also supported this result (Supplementary Fig. 1A, B). We defined the cluster results as Cluster 1 (C1) and Cluster 2 (C2), where C1 showed significantly higher levels of immune cell infiltration compared to C2 (Fig. 1C, D). Therefore, we defined C1 as an immunologically rich tumor and C2 as an immunologically poor tumor. Furthermore, to evaluate the stability and robustness of the clustering results, we used six algorithms to confirm the same findings: The TIMER, quanTlseq, MCP-counter, xCell, EPIC algorithm clustering showed that C1 had higher immune cell abundance than C2 (Supplementary Fig. 1C). The ESTIMATE immune score of C1 was also higher than that of C2 (Fig. 1E).

Construction of WGCNA and identification of immune-related lncRNA modules

To ensure the scale-free network of co-expression, we set the scale-free R^2 to 0.55 and the soft threshold β to 5 (Supplementary Fig. 2D). Using the algorithm (combined dynamic tree cutting), we obtained topologically overlapping clusters, identifying 10 gene expression modules in different colors, and compared and analyzed the correlations between these modules (Supplementary Fig. 1E). We evaluated the expression of each module's correlation with clinical characteristics such as immune clusters, age, TNM stage, Stage, ER, HER2, Status, PR, gender, tumor mutation burden (TMB), and neoantigen load (NAL) (Fig. 1F). The results showed that one module (Immune Cluster) exhibited the most significant correlation with clinical characteristics. In the green module, the correlation coefficient between GS and MM reached 0.96, indicating that the quality of lncRNA module construction is highly reliable (Fig. 1G). lncRNAs in the green module were selected for use in building models based on immune-related lncRNAs.

ImmuLncRNA algorithm screening for lncRNAs related to immunity

We used the ImmuLncRNA algorithm to derive lncRNA regulatory factors involved in immune regulation from lncRNA and gene expression data. From a functional regulation perspective, lncRNA that play a key role in immune regulation may have their target genes specifically enriched at the top or bottom of immune-related pathways. Based on the ImmuLncRNA function, we ultimately identified 1278 immune-related lncRNAs (Supplementary Table 1). We can see that most lncRNA are associated with "Cytokine_Receptors", "Antigen_Processing_and_Presentation", "NaturalKiller_Cell_Cytotoxicity", "TCRsignalingPathway" and "Antimicrobials" (Fig. 1H). Then, we took the intersection of the lncRNAs identified by the ImmuLncRNA algorithm and the hub lncRNAs identified by WGCNA. We obtained 72 immune-related lncRNAs (Fig. 1I).

Construction of a consensus immune-related lncRNA signature

We conducted a univariate Cox regression analysis on the 72 immune-related lncRNAs identified by the ImmuLncRNA algorithm and WGCNA algorithm, revealing that 15 lncRNAs were significant prognostic factors (Supplementary Fig. 1F) for patients. We used a combination of 101 machine learning algorithms to construct a prognostic model (Supplementary Table 2) for these 15 lncRNAs. The results showed that the combination of Lasso regression and StepCox (backward) achieved an optimal average C-Index (Fig. 2A) of 0.71. Subsequently, we performed Lasso regression, using the minimum λ value from Lasso regression to build the prognostic model. We selected non-zero regression coefficient variables as feature screening results and then used StepCox (backward) to construct the model, ultimately identifying 9 lncRNAs significantly associated with prognosis (Fig. 2B–C).

We developed a prognostic risk score model based on these 9 immunoregulatory IRLS, where each patient's risk score is the sum of the product of the expression levels of these 9 immunoregulatory lncRNAs and their corresponding StepCox regression coefficients. The risk score was calculated as follows: IRLS risk score = $(0.215 \times \text{expression of C22orf34}) + (-0.153 \times \text{expression of DBH-AS1}) + (-0.238 \times \text{expression of MIAT}) + (-0.190 \times \text{expression of LINC00892}) + (-0.644 \times \text{expression of PRKAR1B-AS1}) + (0.179 \times \text{expression of PRKCQ-AS1}) + (-0.196 \times \text{expression of MIR3142HG}) + (0.205 \times \text{expression of CARD8-AS1}) + (-0.494 \times \text{expression of LINC01943})$. Notably, PRKAR1B-AS1 exhibited the strongest association with the IRLS score. We determined the optimal cut-off values for high and low risk groups using the surv_cutpoint and surv_categorize algorithms. Subsequently, we performed survival analysis on the TCGA-BRCA dataset based on the optimal cut-off values for high and low risks. The results showed that patients in the high-risk group had significantly lower OS ($P < 0.05$), GSE58812, GSE103091, GSE42568, GSE20711, GSE16446, GSE20713, and Meta-Cohort data also demonstrated the same trend, with OS in the high-risk group being significantly lower than in the low-risk group ($P < 0.05$) (Fig. 2D–K). These findings confirm the robustness and reliability of our model. Subsequently, the optimal cutoff values for stratifying PRKAR1B-AS1 expression levels was determined using the surv_cutpoint and surv_categorize algorithms, categorizing patients into high- and low-expression groups. Patients with high PRKAR1B-AS1 expression consistently demonstrated significantly prolonged OS ($P < 0.05$) across all validation cohorts, including TCGA-BRCA, GSE58812, GSE103091, GSE42568, GSE20711, GSE16446, GSE20713, and the Meta-Cohort dataset (Supplementary Fig. 2). Additionally, to explore the impact of clinical characteristics, including IRLS, on patient OS, we conducted a multivariate Cox regression analysis on the aforementioned dataset, which revealed that IRLS is an independent prognostic factor for OS in breast cancer patients.

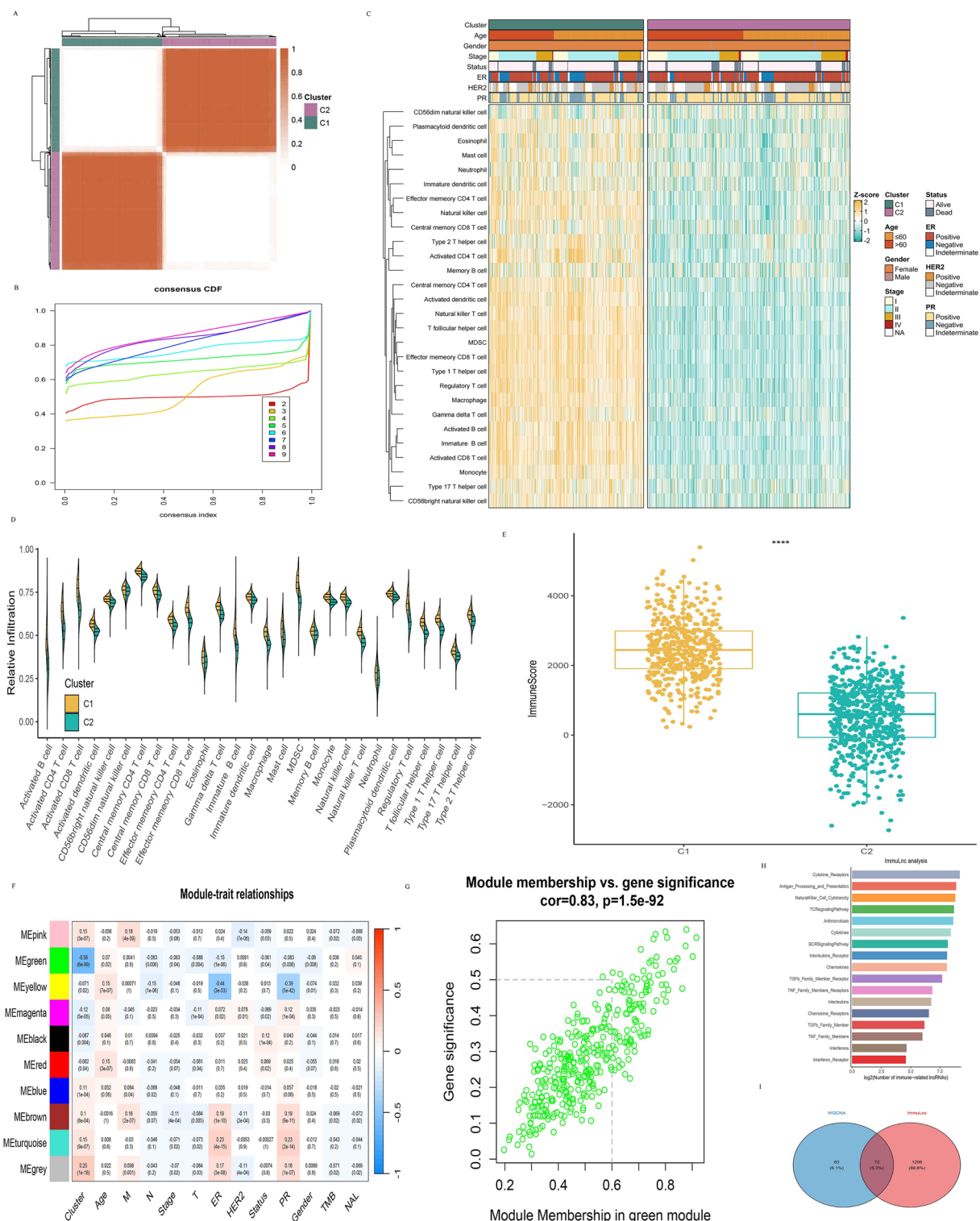
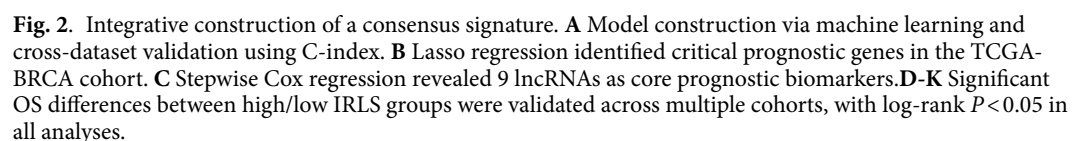


Fig. 1. Identification of immune-related lncRNAs. **A** Consensus clustering scores (k=2); **B** Optimal k-value selection; **C–D** Immune cell infiltration analysis between clusters (ssGSEA); **E** Differential ESTIMATE immune scores (TCGA-BRCA cohort, **** $P < 0.0001$); **F** Module-clinical trait correlation; **G** High-GS/MM lncRNAs in the green module; **H** ImmLnc-identified lncRNAs linked to immune pathways; **I** Overlapping lncRNAs from WGCNA and ImmLnc.



These clinical characteristics include age, TNM stage, stage, ER, PR, HER2, TMB, NAL, gender, grade, and subtype (Supplementary Fig. 3). Subsequently, we further explored the predictive performance of IRLS on DFS in 11 (TCGA-BRCA, GSE58812, GSE103091, GSE42568, GSE20711, GSE16446, GSE20713, GSE9195, GSE20685, GSE21653, GSE61304) datasets, which also showed that patients in the high-risk group had significantly worse DFS. Meta-Cohort also demonstrated the same result ($P < 0.05$). However, no significant difference in DFS was observed between high-risk and low-risk groups in the GSE9195 and GSE20685 datasets (Supplementary Fig. 4). This may be related to the deviations caused by the smaller sample sizes in these two datasets. In addition, we explored the effect of IRLS and clinical characteristics on DFS in 11 data sets. Multivariate Cox regression results showed that IRLS was an independent prognostic factor for DFS in TCGA-BRCA, GSE103091 and GSE20711 data sets, but IRLS was not observed to have a significant effect on DFS in other data sets (Supplementary Fig. 5).

Evaluation of the IRLS model

In order to verify the predictive performance of our IRLS model for breast cancer patient survival prognosis, we used a variety of algorithms to calculate different indicators of the model. We used the timeROC algorithm to calculate the AUC value of 1-, 3-, and 5-year OS in 8 data sets.

The AUC values for TCGA-BRCA at 1, 3, and 5 years were 0.782, 0.776, and 0.744, respectively; 0.811, 0.758, and 0.792 in GSE58812; 0.832, 0.778, and 0.692 in GSE103091; 0.793, 0.796, and 0.776 in GSE42568; 0.741, 0.732, and 0.675 in GSE20711; 0.753, 0.721, and 0.653 in GSE16446; 0.805, 0.794, and 0.627 in GSE20713; and 0.775, 0.754, and 0.649 in Meta-Cohort (Fig. 3A, Supplementary Table 3). The C-index values were 0.751, 0.737, 0.707, 0.743, 0.698, 0.673, 0.708, and 0.747, respectively (Fig. 3B, Supplementary Table 3). The IBS values were 0.109, 0.112, 0.1

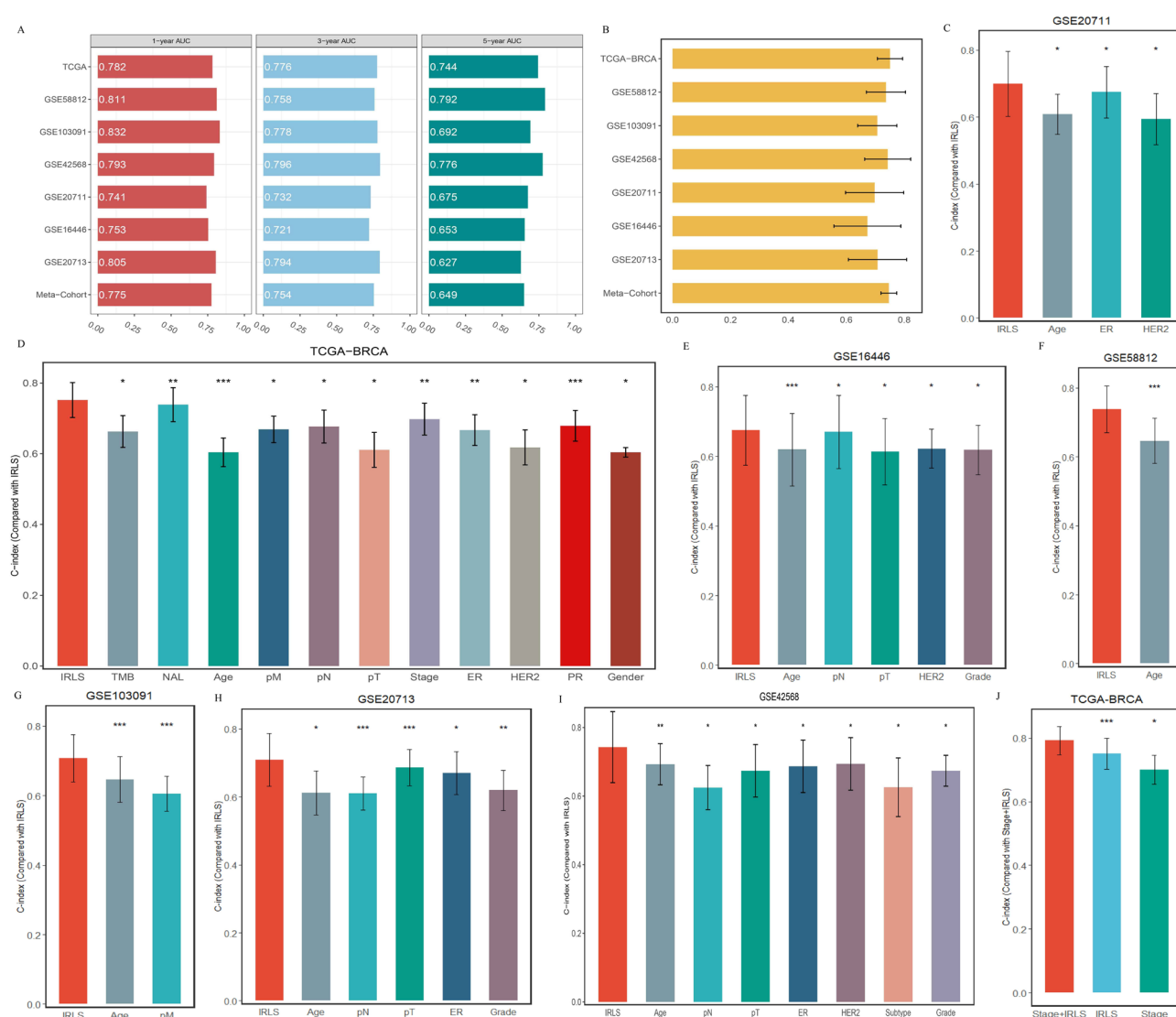


Fig. 3. IRLS model evaluation. **A** The AUC of 1-, 3-, 5-year OS prediction. **B** C-index of eight datasets. **C–I** Prognostic performance comparison with clinical variables. **J** Prognostic performance comparison of IRLS + Stage vs. IRLS and AJCC stage alone in TCGA-BRCA ($n = 1108$). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

B

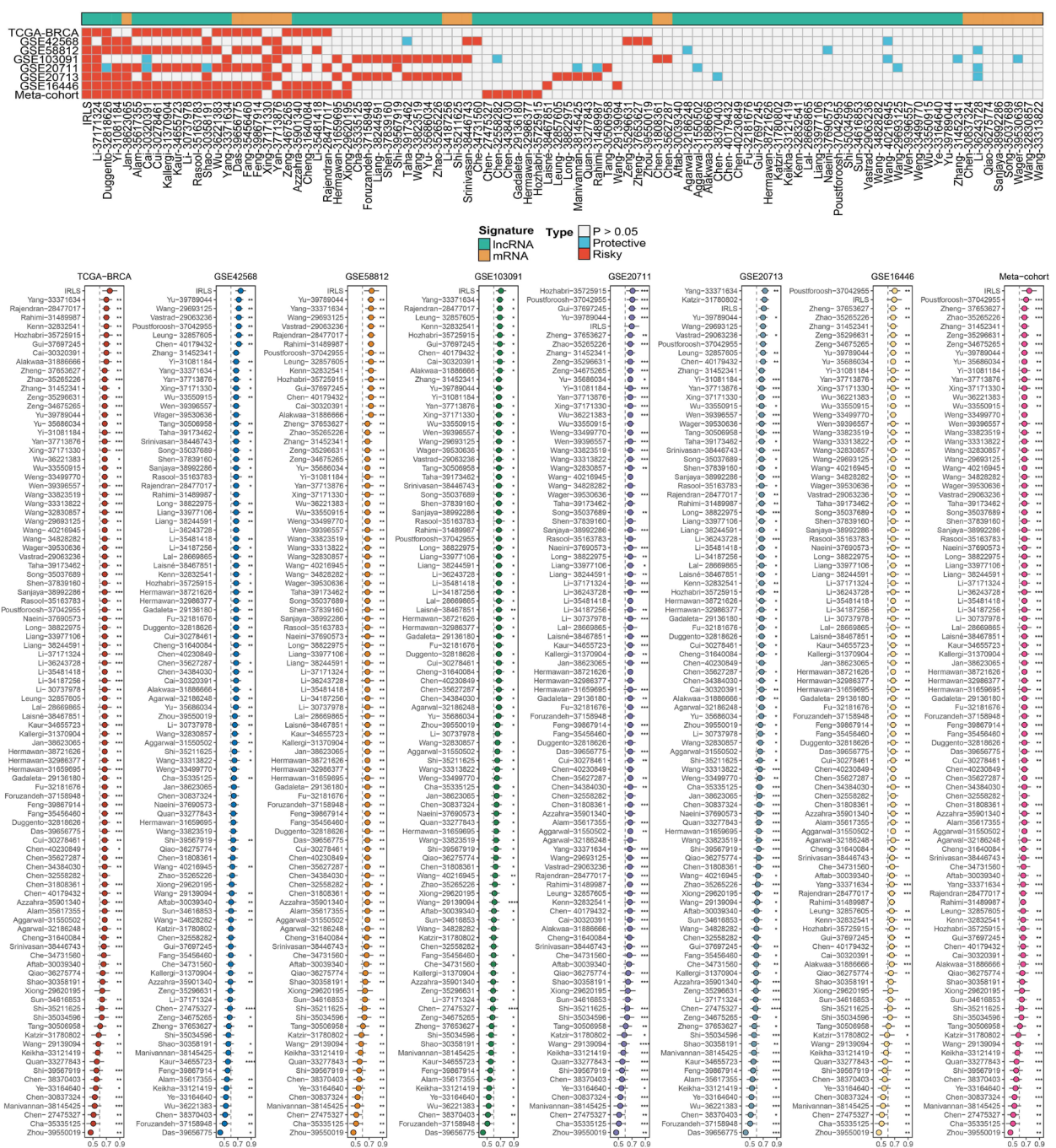


Fig. 4. Prognostic model evaluation in breast cancer cohorts. **A** Univariate Cox analysis of IRLS and 95 published models in TCGA-BRCA ($n=1108$), GSE42568 ($n=88$), GSE58812 ($n=107$), GSE103091 ($n=238$), GSE20711 ($n=88$), GSE20713 ($n=104$), GSE16446 ($n=102$) and Meta-cohort (1835). **B** C-index comparison of IRLS and 95 published models in TCGA-BRCA, GSE42568, GSE58812, GSE103091, GSE20711, GSE20713, GSE16446 and Meta-cohort. $^*P<0.05$; $^{**}P<0.01$; $^{***}P<0.001$; $^{****}P<0.0001$.

53,0.116,0.107,0.111,0.127, and 0.115, and the iAUC values were 0.872,0.943,0.873,0.802,0.782,0.754,0.862, and 0.756 (Supplementary Fig. 6, Supplementary Table 3). The values of these indicators show that our constructed IRLS Model has high reliability and distinctiveness in predicting OS for breast cancer patients. Subsequently, we evaluated the prognostic performance of PRKAR1B-AS1 expression in breast cancer patients. Using the timeROC algorithm, we calculated AUC values for 1-, 3-, and 5-year OS across eight datasets, while the survival algorithm was employed to compute C-index values. The results demonstrated robust predictive power of PRKAR1B-AS1 for patient outcomes (Supplementary Table 4), with high concordance between its predictive capacity and that of the IRLS model. Additionally, many studies have explored the predictive performance of clinical characteristics (such as age, TNM stage, Stage, ER, PR, HER2, TMB, NAL, gender, grade, and subtype)

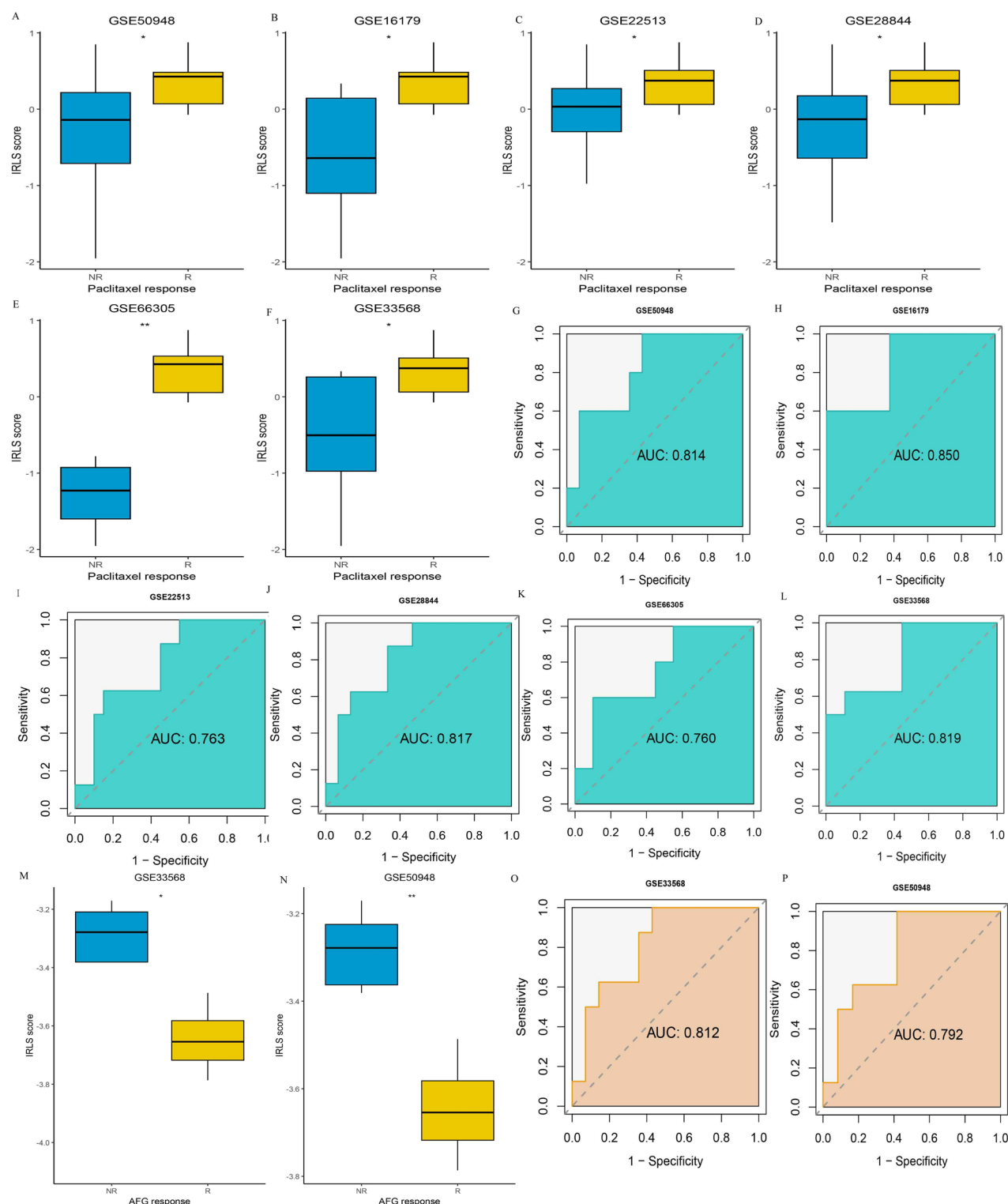


Fig. 5. Predictive value of IRLS for Paclitaxel and AFG response. **A–F** IRLS score distribution between responders and nonresponders of Paclitaxel in six datasets. **G–L** ROC curves of IRLS to predict the benefits of Paclitaxel in six datasets. **M–N** IRLS score distribution between responders and nonresponders of anastrozole-fulvestrant-gefitinib (AFG) in two datasets. **O–P** ROC curves of IRLS to predict the benefits of AFG in two datasets. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

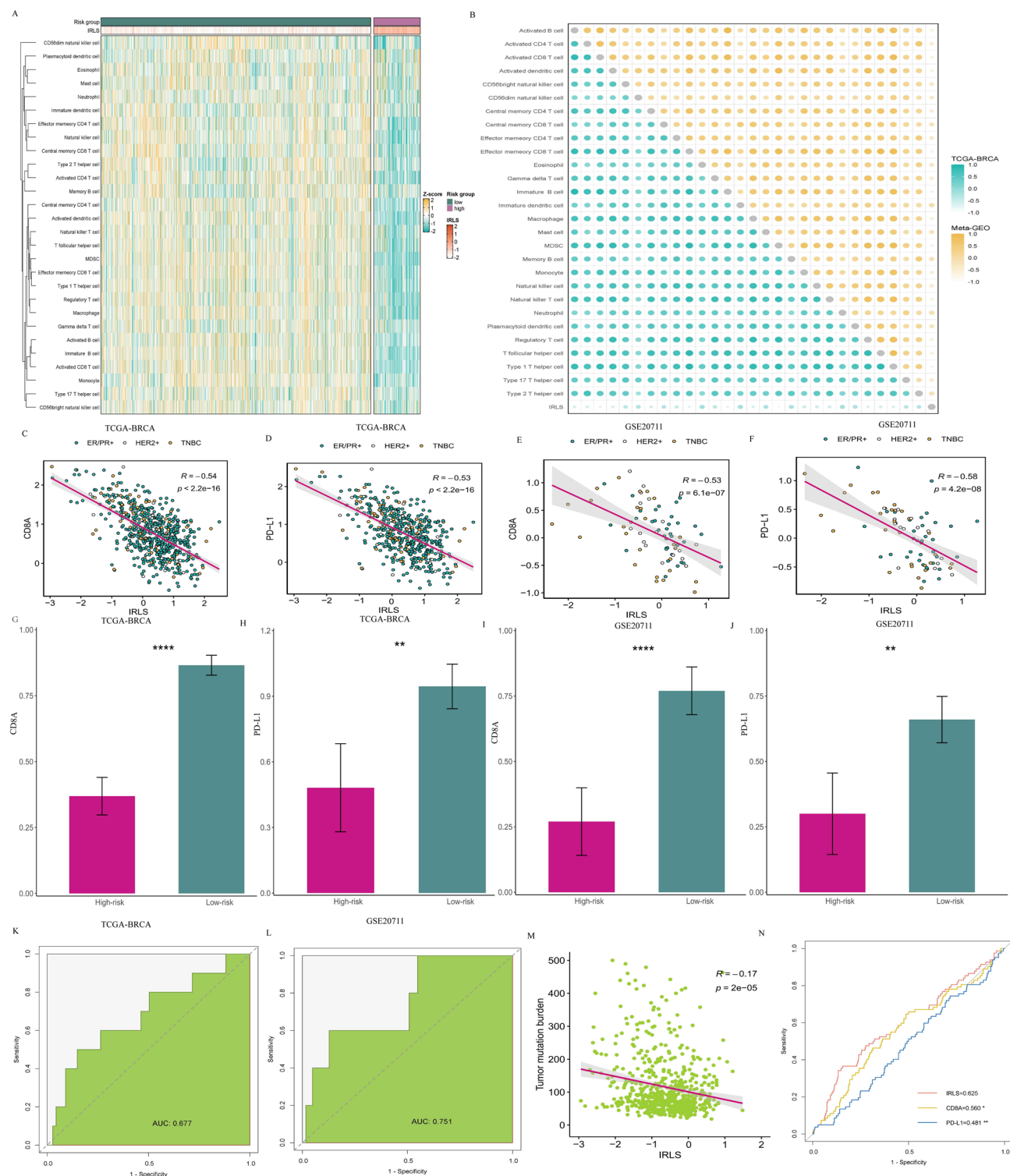


Fig. 6. Validation of IRLS-ICI associations. **A** The relationship between IRLS and immune infiltration in TCGA-BRCA. **B** Correlation heatmap of Pearson's correlations (IRLS and immune infiltration) in TCGA-BRCA and Meta-GEO. **C–D** IRLS-CD8A correlations in TCGA-BRCA. **E–F** IRLS-PD-L1 correlations in GSE20711. **G–H** Intergroup CD8A and PD-L1 differences in TCGA-BRCA. **I–J** Intergroup CD8A and PD-L1 differences in GSE20711. **K–L** ROC curves of IRLS for molecular expression prediction in TCGA-BRCA and GSE20711. **M** Correlation between IRLS and TMB. **N** ROC analysis (IRLS/CD8A/PD-L1) for atezolizumab response prediction. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

on OS, and our study also delved into this area. We compared the predictive performance of these indicators on survival outcomes with the IRLS model, and the results showed that our IRLS model had a higher C-index and was significantly different ($P < 0.05$) (Fig. 3C-I). AJCC stage plays a central role in the diagnosis and treatment of breast cancer patients, providing scientific evidence for clinical decision-making, prognosis assessment, and treatment optimization. To explore whether the combination of IRLS model and AJCC stage is superior to either IRLS or Stage alone, we used the combined IRLS model and AJCC stage as a benchmark, calculated the C-index values for different indicators, and found that the combined AJCC stage and IRLS indicators had the highest C-index at 0.792, with significant differences compared to the other two individual indicators ($P < 0.05$) (Fig. 3J).

Assessment of gene expression-derived predictive models for breast cancer

To verify the predictive performance of our constructed IRLS model compared to other models that have been published, we retrieved and collected 95 gene expression prediction models (Supplementary Table 5). These gene expression markers are involved in various biological processes. The results of univariate Cox regression suggest that among all models, only IRLS showed significant differences in predicting OS for breast cancer patients across eight datasets, with lower IRLS scores associated with longer OS (Fig. 4A). Subsequently, we evaluated the C-index of all models across the eight datasets and compared it with the C-index of IRLS. We found that IRLS had the highest C-index in the TCGA-BRCA, GSE42568, GSE58812, GSE103091 and Meta-cohort datasets, showing significant differences from most other models ($P < 0.05$). In the GSE20711, GSE20713 and GSE16446 datasets, IRLS also demonstrated excellent predictive performance, showing significant differences compared to other models ($P < 0.05$) (Fig. 4B). It can be observed that other models exhibit poorer stability compared to the IRLS model, performing well only on their own training sets or a few other datasets, while performing poorly on other validation datasets. This may be related to our use of multiple machine learning algorithms to build the models.

Predictive value of IRLS score for different treatment regimens

Previous studies have suggested that lncRNA may be associated with the resistance to chemotherapy and targeted therapy in breast cancer patients. To explore this, we investigated the predictive value of IRLS scores for different treatment outcomes in breast cancer patients. Six datasets based on Paclitaxel treatment were included. A total of 112 eligible patients were enrolled, with 61 responding to Paclitaxel and 51 not responding. The results showed that the IRLS score for patients who responded to Paclitaxel was significantly higher than those who did not ($P < 0.05$) (Fig. 5A-F). To further evaluate the predictive power of our model, we conducted ROC analysis, which revealed area under the curve values of 0.814, 0.850, 0.763, 0.817, 0.760, and 0.819 for the GSE50948, GSE16179, GSE22513, GSE28844, GSE66305 and GSE33568 datasets (Fig. 5G-L). This indicates that IRLS has high discriminative value in predicting the response to Paclitaxel treatment in patients. In addition, we explored the differences in IRLS scores between responders and non-responders in the anastrozole-fulvestrant-gefitinib (AFG) combination therapy in the GSE33568 and GSE50948 datasets. The results showed that the IRLS score for responders to AFG was significantly lower than that of non-responders ($P < 0.05$) (Fig. 5M-N). The area under the curve for the GSE33568 and GSE50948 datasets was 0.812 and 0.792, respectively, which also indicates that the IRLS score has high predictive value in predicting AFG regimen responsiveness (Fig. 5O-P).

Impact of IRLS on ICI therapy

We applied the ssGSEA algorithm to quantify immune scores in patients from the TCGA-BRCA and Meta-GEO cohorts. The results revealed that low-risk group patients exhibited significantly higher immune cell infiltration (Fig. 6A, Supplementary Fig. 7A), while a negative correlation was observed between IRLS scores and immune cell infiltration levels (Fig. 6B). In the TCGA-BRCA dataset, we found that as the IRLS score increased, the expression levels of CD8A and PD-L1 decreased (Fig. 6C, D, Supplementary Fig. 7B, C). A similar trend was observed in the GSE20711 dataset ($P < 0.05$) (Fig. 6E, F). We found that the expression levels of CD8A and PD-L1 were higher in the low-risk group compared to the high-risk group (Fig. 6G-J). Additionally, we discovered that IRLS can predict molecular expression in breast cancer patients, with lower IRLS scores in TNBC patients (Supplementary Fig. 7D, E) ($P < 0.05$). Subsequently, we evaluated the predictive power of IRLS for molecular expression in breast cancer patients using ROC curves, which showed an AUC value of 0.677 in TCGA-BRCA and 0.751 in GSE20711 (Fig. 6K, L). We used the TCGA-BRCA dataset to evaluate the relationship between IRLS and TMB or NAL. The results showed that there was a significant negative correlation between IRLS scores and TMB or NAL ($P < 0.05$) (Fig. 6M, Supplementary Fig. 7F). Patients with lower IRLS scores may have higher genetic instability. Additionally, we assessed the predictive performance of IRLS scores on atezolizumab treatment response using the GSE157284 dataset. The results indicated that compared to CD8A and PD-L1, IRLS scores achieved the highest AUC value of 0.625, suggesting that IRLS has some predictive value for the efficacy of atezolizumab therapy (Fig. 6N).

Discussion

Breast cancer is the most common malignant tumor among women worldwide, and drug resistance and recurrence metastasis are key factors contributing to poor prognosis in breast cancer patients²¹. Although early diagnosis of breast cancer can improve survival rates, it is often only discovered or diagnosed after metastasis has occurred¹. The status of lymph nodes, tumor size, and histological grade are three major prognostic determinants used in routine practice²². However, breast cancer exhibits high heterogeneity, meaning that even patients with identical clinical and pathological features may have significantly different prognoses. Prognostic and predictive factors are essential tools for personalized breast cancer treatment. Developing new predictive methods for breast cancer is highly necessary, as they facilitate clinical decision-making and therapeutic guidance. Recent deep sequencing studies of the transcriptome have revealed that approximately 4/5 of human genome transcripts

are non-coding genes, including lncRNA²³. lncRNA has been shown to participate in various aspects of breast cancer development, progression, invasion, and metastasis^{24,25}, and lncRNA with different phenotypes are increasingly being used to construct prognostic features^{26–28}. However, there is less research on immune-related lncRNA and their association with breast cancer prognosis. To fill this gap, we developed the IRLS score to evaluate its predictive performance in breast cancer prognosis, recurrence, and therapeutic response to both chemotherapy and immunotherapy agents.

We first calculated the immune cell infiltration scores for samples in the TCGA-BRCA dataset using the ssGSEA algorithm, followed by unsupervised consensus clustering of these scores. Based on WGCNA analysis, we identified candidate lncRNA related to immunity. Additionally, we used the ImmuLncRNA algorithm to derive lncRNA regulatory factors involved in immune regulation. We intersected the hub lncRNAs identified by WGCNA with the lncRNAs identified by the ImmuLncRNA algorithm, resulting in 72 immune-related lncRNAs. Subsequently, we screened prognostic models using a combination of 101 machine learning algorithms for lncRNAs that influence patient prognosis. Finally, based on Lasso regression and StepCox (backward), we identified 9 lncRNAs significantly associated with prognosis to develop an IRLS score model. Using this model, we found that IRLS is an independent prognostic factor for OS in breast cancer patients, and patients with higher IRLS scores have significantly poorer OS. IRLS also has some predictive value for DFS but is not an independent predictor of DFS. We then calculated the AUC and C-Index values for the IRLS score model across seven datasets and the Meta-Cohort cohort, demonstrating its strong performance in predicting patient outcomes. Based on PRKAR1B-AS1 expression levels, breast cancer patients were stratified into high- and low-expression groups. Patients with elevated PRKAR1B-AS1 expression demonstrated significantly improved OS ($P < 0.05$). This finding aligns with Saidy et al.'s report showing enhanced disease-specific survival in breast cancer patients with high PRKAR1B-AS1 expression²⁹. Furthermore, the prognostic predictive power of PRKAR1B-AS1 expression exhibited high concordance with the IRLS model. We also found that compared to other clinical factors predicting patient outcomes, IRLS has a higher C-Index with statistically significant differences. The combination of IRLS and AJCC stage may offer better predictive performance. We compared the IRLS model constructed in this study with 95 prognostic markers selected from other studies using univariate Cox regression analysis, and found that only the IRLS model showed significant differences across all datasets. We observed that the IRLS score demonstrated relatively high predictive performance in various datasets. Some models, such as (Yang-33371634, Rajendran-28477017), have shown good predictive capabilities in their own research models or a few external models, which may be related to their validation being limited to internal models^{30,31}. We used multiple machine learning algorithms to obtain IRLS scores, which showed higher predictive power and extrapolation ability.

Paclitaxel is a cornerstone drug in breast cancer chemotherapy. To explore the performance of IRLS in guiding clinical treatment, we collected six GEO datasets, which showed that patients responding to Paclitaxel had higher IRLS scores. Previous studies have indicated that lncRNA may influence patient sensitivity to paclitaxel treatment by activating certain pathways^{32,33}. Additionally, we evaluated the predictive utility of the IRLS for therapeutic response to AFG combination therapy in breast cancer. Notably, AFG responders demonstrated significantly lower IRLS scores compared to non-responders. Therefore, IRLS can provide some predictive value for assessing the effectiveness of chemotherapy regimens based on Paclitaxel or AFG.

In recent years, immunotherapy has emerged as a promising treatment strategy in the field of breast cancer, particularly with immune checkpoint inhibitors targeting PD-L1 receiving significant attention. ICI demonstrates unique therapeutic advantages by reactivating the body's anti-tumor immune function. Currently, research on breast cancer immunotherapy has covered various molecular subtypes, especially achieving important breakthroughs in TNBC³⁴. The Impassion-130 study evaluated the efficacy of atezolizumab combined with albumin-bound paclitaxel in untreated advanced TNBC patients. The results showed that in the intention to treat (ITT) population, compared to the placebo plus albumin-bound paclitaxel group, atezolizumab plus albumin-bound paclitaxel extended PFS (7.2 months vs. 5.5 months, HR = 0.80, 95% CI: 0.69 ~ 0.92, $P = 0.002$) and OS (25 months vs. 15.5 months, HR = 0.62, 95% CI: 0.45 ~ 0.86)^{35,36}. Subsequently, we explored the predictive performance of IRLS for immunotherapy benefits. The results showed that the low-risk group classified by IRLS had more immune cell infiltration, and there was a significant negative correlation between CD8A and PD-L1 with IRLS scores. Low-risk patients exhibited higher expressions of CD8A and PD-L1. We also found that TNBC patients had lower IRLS scores, suggesting that TNBC may benefit more from immunotherapy compared to other types of breast cancer, which is consistent with previous studies^{37,38}. Additionally, we observed a significant negative correlation between IRLS scores and TMB and NAL, indicating that patients with low IRLS scores have greater tumor immunogenicity, thus triggering an immune response against the tumor. These results suggest that IRLS scores may be used to predict the efficacy of ICI in treating breast cancer. Therefore, IRLS scores could serve as a strong biomarker for predicting the effectiveness of ICI therapy. Breast cancer patients with lower IRLS scores may benefit more from ICI treatment. We also evaluated the predictive role of IRLS scores in the benefits of atezolizumab treatment for breast cancer. The results showed that IRLS scores have higher predictive performance compared to CD8A and PD-L1. In summary, the IRLS model enhances clinical prognostication by complementing established frameworks like AJCC staging. Its integration with routine molecular profiling (e.g., ER/PR/HER2 status) could refine risk stratification, identifying high-risk patients for intensified surveillance or adjuvant therapy. Crucially, IRLS predicts differential treatment responses: low-risk patients show enhanced immune infiltration and may benefit from ICIs (e.g., atezolizumab), while high-risk patients exhibit chemosensitivity to paclitaxel-based regimens. For implementation, a standardized PCR-based assay targeting the 9-lncRNA signature could be deployed alongside multigene tests, enabling personalized therapeutic decisions in both early and advanced settings.

While the IRLS model demonstrates robust clinical utility across 17 cohorts, further experimental investigation of key lncRNAs (e.g., PRKAR1B-AS1) is deferred due to current resource constraints. However,

the IRLS score demonstrated significant negative correlations with immune effectors (CD8A, PD-L1) while showing good predictive performance for atezolizumab response (AUC=0.625). These findings provide a basis for subsequent functional studies to elucidate the roles of these lncRNAs in paclitaxel sensitivity and immune checkpoint regulation.

Conclusion

In summary, we constructed the IRLS model using machine learning and various algorithms based on a large number of public datasets to evaluate the prognosis, recurrence, and response to chemotherapy and immunotherapy in breast cancer patients. The IRLS scores exhibit robust predictive performance and extrapolation across different datasets. This is a promising tool for guiding clinical treatment and can be used to inform personalized therapy for breast cancer patients.

Data availability

In this study, we utilized GEO datasets including GSE42568, GSE58812, GSE103091, GSE20711, GSE20713, GSE16446, GSE9195, GSE20685, GSE21653, GSE61304, GSE50948, GSE16179, GSE22513, GSE28844, GSE66305, GSE33568, and GSE157284. All samples and files were supported by the GEO (<https://www.ncbi.nlm.nih.gov/geo>) and TCGA databases (<http://www.cancer.gov/tcga>).

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Author contributions

Y.X.L., S.S.C., D.F.Y. and J.T.C. designed the study and wrote the manuscript. S.S.C., J.T.C. and D.F.Y. analyzed data and contributed to writing the manuscript. C.M.L., C.H.T., S.S.C. revised the first draft. Y.X.H. and S.S.C. critically revised the manuscript and provided final approval. Y.X.L., J.T.C. and D.F.Y. contributed equally to this work. Y.X.H. and S.S.C. are co-corresponding authors. All authors read the manuscript and approved its content.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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